

Project P4: Brute-Force Detection & Investigation

Authentication Telemetry Pipeline, Multi-Account Spray Analysis & Operational SOC Dashboarding

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SOC-REP-P4-2026-0913	Natto Chakma	September 13, 2026	TLP:CLEAR
TARGET MONITORED HOST	ATTACKING SOURCE IP	SPLUNK ARCHITECTURE	INGESTION INDEX
ubuntu-p3 (192.168.100.9)	192.168.100.7 (wazuh-server)	Splunk Enterprise 10.4.3	index=linux_security

1. Executive Summary

During routine operational threat monitoring within the Splunk SOC laboratory network, an automated alert detected abnormal authentication volume against the monitored Linux endpoint **ubuntu-p3** (192.168.100.9). An adversary located at internal IP **192.168.100.7** initiated an automated credential guessing and password spray campaign over SSH (TCP 22).

Key Incident Finding: A total of **13 authentication failures** were generated within a 3-minute window (00:13:12 to 00:16:15). Forensic analysis isolated a dual-vector attack: a **horizontal password spray** probing 8 distinct usernames (admin, root, guest, test, user1, user2, service, backup) and a **targeted brute-force burst** focusing on privileged account admin (4 attempts). Legitimate administrative access was confirmed for user natto from host 192.168.100.1. No breach occurred.

2. Laboratory Architecture & Data Pipeline

The monitoring architecture spans two virtual machines on VirtualBox isolated network LabNetwork (192.168.100.0/24):

System Node	Network IP	Operating System	Role in Telemetry Pipeline
ubuntu-p3	192.168.100.9	Ubuntu Server 24.04.5 LTS	Monitored endpoint hosting SSH daemon and Splunk Universal Forwarder 10.4.3.
wazuh-server	192.168.100.7	Ubuntu Server 24.04.4 LTS	Central SIEM hosting Splunk Enterprise 10.4.3 receiver (TCP 9997) and Search Head.
Lab Gateway	192.168.100.1	VirtualBox Gateway	Administrative jump host connecting via host loopback port forwarding.

3. Telemetry Ingestion & Field Normalization

The Splunk Universal Forwarder on ubuntu-p3 continuously monitors /var/log/auth.log, transporting raw authentication strings to index=linux_security under sourcetype linux_secure. Because raw logs contain unindexed variables, a normalized SPL regular expression was validated to extract standard security entities:

```
index=linux_security sourcetype=linux_secure ("Failed password" OR "Accepted")
| rex "Failed password for (invalid user )?(?<username>\S+) from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"
| rex "Accepted \S+ for (?<username>\S+) from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"
| eval auth_status=if(match(_raw, "Accepted"), "SUCCESS", "FAILURE")
| table _time host src_ip username auth_status _raw | sort - _time
```

4. Forensic Evidence Collection & Analysis

The query successfully indexed and parsed **14 events** within the investigation window, segregating attacker actions from legitimate operations:

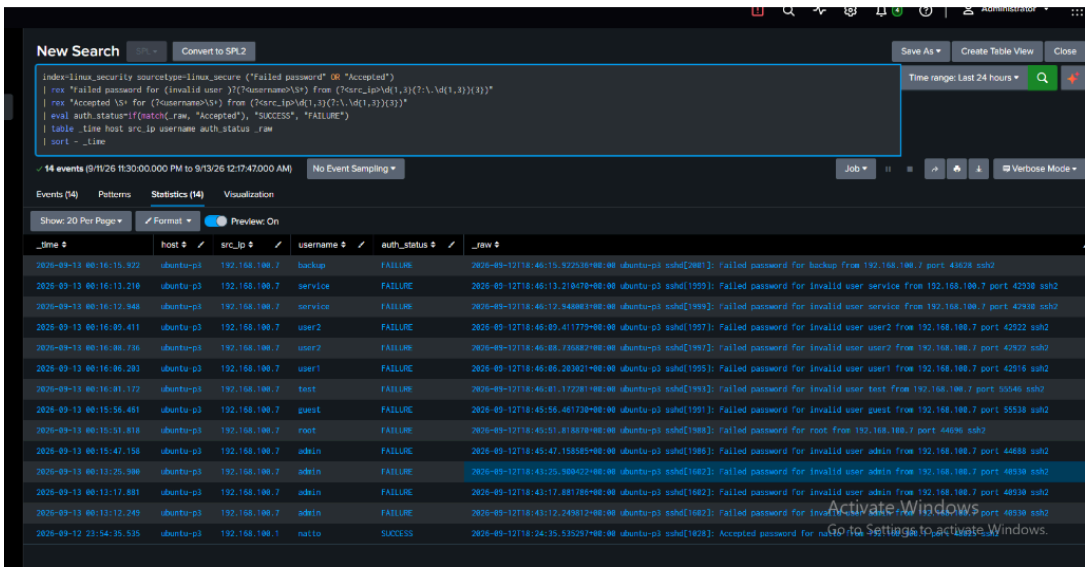


Exhibit EX-P4-01: Live Splunk Search & Reporting view confirming 14 parsed authentication records, separating 13 failures from 1 legitimate success.

Chronological Incident Timeline (14 Verified Events)

Timestamp (UTC+5:30)	Host	Source IP	Target Account	Result	Raw Telemetry / Error Event Details
2026-09-13 00:16:15	ubuntu-p3	192.168.100.7	backup	FAILURE	Failed password for backup from 192.168.100.7 port 43628 ssh2
2026-09-13 00:16:13	ubuntu-p3	192.168.100.7	service	FAILURE	Failed password for invalid user service from 192.168.100.7 port 42930 ssh2
2026-09-13 00:16:12	ubuntu-p3	192.168.100.7	service	FAILURE	Failed password for invalid user service from 192.168.100.7 port 42930 ssh2
2026-09-13 00:16:09	ubuntu-p3	192.168.100.7	user2	FAILURE	Failed password for invalid user user2 from 192.168.100.7 port 42922 ssh2
2026-09-13 00:16:08	ubuntu-p3	192.168.100.7	user2	FAILURE	Failed password for invalid user user2 from 192.168.100.7 port 42922 ssh2
2026-09-13 00:16:06	ubuntu-p3	192.168.100.7	user1	FAILURE	Failed password for invalid user user1 from 192.168.100.7 port 42916 ssh2
2026-09-13 00:16:01	ubuntu-p3	192.168.100.7	test	FAILURE	Failed password for invalid user test from 192.168.100.7 port 55546 ssh2
2026-09-13 00:15:56	ubuntu-p3	192.168.100.7	guest	FAILURE	Failed password for invalid user guest from 192.168.100.7 port 55538 ssh2
2026-09-13 00:15:51	ubuntu-p3	192.168.100.7	root	FAILURE	Failed password for root from 192.168.100.7 port 44696 ssh2
2026-09-13 00:15:47	ubuntu-p3	192.168.100.7	admin	FAILURE	Failed password for invalid user admin from 192.168.100.7 port 44688 ssh2
2026-09-13 00:13:25	ubuntu-p3	192.168.100.7	admin	FAILURE	Failed password for invalid user admin from 192.168.100.7 port 40930 ssh2
2026-09-13 00:13:17	ubuntu-p3	192.168.100.7	admin	FAILURE	Failed password for invalid user admin from 192.168.100.7 port 40930 ssh2
2026-09-13 00:13:12	ubuntu-p3	192.168.100.7	admin	FAILURE	Failed password for invalid user admin from 192.168.100.7 port 40930 ssh2

Timestamp (UTC+5:30)	Host	Source IP	Target Account	Result	Raw Telemetry / Error Event Details
2026-09-12 23:54:35	ubuntu- p3	192.168.100.1	natto	SUCCESS	Accepted password for natto from 192.168.100.1 port 48025 ssh2

5. Operational SOC Dashboard: Brute-Force Detection & Investigation

To provide tier-1 and tier-2 SOC analysts with immediate situational awareness during authentication attacks, a dedicated dark-mode operational dashboard was designed and deployed in Splunk Web.

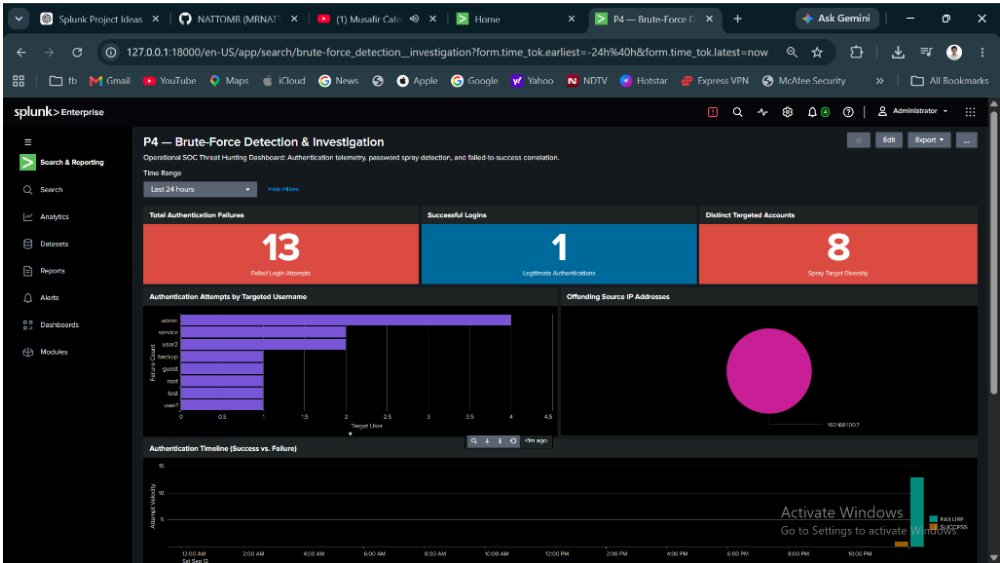


Exhibit EX-P4-02: Live view of the operational "P4 — Brute-Force Detection & Investigation" SOC dashboard in Splunk Enterprise.

Operational Dashboard Panel Breakdown

#	Panel Name	Visualization Type	Observed Lab Metric	SOC Analyst Operational Purpose
01	Total Authentication Failures	Single Value (Red Block)	13 attempts	High-priority metric triggering when failure counts exceed baseline thresholds.
02	Successful Logins	Single Value (Blue Block)	1 authentication	Tracks legitimate user logins to correlate whether credential guessing was successful.
03	Distinct Targeted Accounts	Single Value (Red Block)	8 distinct users	Distinguishes single-account lockouts from horizontal multi-account password spraying.
04	Attempts by Targeted Username	Horizontal Bar Chart	admin (4), service (2), user2 (2), etc.	Visualizes distribution across privileged accounts (admin/root) vs service accounts.
05	Offending Source IP Addresses	Pie Chart	100% (192.168.100.7)	Identifies primary attacking IP addresses for immediate firewall isolation.
06	Authentication Timeline	Stacked Column Timechart	Burst: 00:13:00 to 00:16:15	Shows attack velocity over time; differentiates human typing from automated scripts.
07	Live Authentication Stream	Tabular Event Feed	14 parsed records	Real-time forensic feed showing precise timestamps, target accounts, and source IPs.

6. Production SPL Detection Query Library

Five specialized SPL queries were engineered and empirically verified against the live lab dataset:

- **High-Volume Failure Velocity (5-min window):**

```
index=linux_security sourcetype=linux_secure "Failed password"
| rex "Failed password for (invalid user )?(?<username>\S+) from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"
| bin _time span=5m | stats count as failures dc(username) as targets by _time, src_ip, host
| where failures >= 5 | sort - failures
```

- **Horizontal Password Spray Detector:**

```
index=linux_security sourcetype=linux_secure "Failed password"
| rex "Failed password for (invalid user )?(?<username>\S+) from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"
| stats count as total_failures dc(username) as unique_targets values(username) as targets by src_ip
| eval attack_type = if(unique_targets >= 3, "Horizontal Password Spray", "Targeted Brute-Force")
| table src_ip attack_type total_failures unique_targets targets | sort - total_failures
```

- **Failed-to-Success Outcome Correlation:**

```
index=linux_security sourcetype=linux_secure ("Failed password" OR "Accepted")
| rex "Failed password for (invalid user )?(?<username>\S+) from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"
| rex "Accepted \S+ for (?<username>\S+) from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"
| eval is_fail=if(match(_raw, "Failed password"), 1, 0), is_succ=if(match(_raw, "Accepted"), 1, 0)
| stats sum(is_fail) as failures, sum(is_succ) as successes by src_ip, username
| eval alert_severity=case(failures >= 3 AND successes >= 1, "CRITICAL: Potential Account Breach", failures >= 5, "HIGH: Active Brute-Force", true(), "LOW")
| table alert_severity src_ip username failures successes | sort - failures
```

7. MITRE ATT&CK Matrix Mapping

Tactic	Technique Name	Technique ID	Observed Adversary Execution & Detection Rule
Credential Access	Brute Force	T1110	Automated script submitting repeated SSH authentication attempts in rapid succession.
Credential Access	Password Guessing	T1110.001	Repeated failed attempts targeting single administrative account admin (4 failures).
Credential Access	Password Spraying	T1110.003	Probing 8 distinct accounts (admin, root, guest, test, user1, user2, service, backup) from 192.168.100.7.
Initial Access	Valid Accounts	T1078	Monitored via failed-to-success correlation to verify whether valid access was achieved.

8. Incident Remediation & Defensive Hardening

1. **Host-Level Automatic Banning (fail2ban):** Deploy fail2ban on ubuntu-p3 to temporarily drop source IPs exceeding 5 failed login attempts in 10 minutes:

```
sudo apt-get install -y fail2ban
sudo systemctl enable --now fail2ban
```

2. **Enforce Key-Based Authentication:** Disallow password authentication in /etc/ssh/sshd_config:

```
PasswordAuthentication no
PubkeyAuthentication yes
```

3. **UFW Network Filtering:** Restrict SSH management access exclusively to authorized jump-host IP 192.168.100.1:

```
sudo ufw allow from 192.168.100.1 to any port 22 proto tcp
sudo ufw deny 22/tcp
```